

Financial Analysis Report

Mobile Banking Application

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1. Executive Summary

This report evaluates the financial viability and risk implications of scope changes in the Mobile Banking Application project. The inclusion of additional security features—mandated by regulatory compliance—resulted in an increase in total project cost from \$500,000 to \$600,000 and extended the schedule from 12 to 14 months.

The original ROI target was 25% over two years, but the revised ROI is -20%, with a Cost-Benefit Ratio (CBR) of 0.83. Despite these negative short-term financial indicators, the scope change is essential to mitigate compliance risks and safeguard long-term trust in the application.

This report applies key project evaluation techniques including EAC, ETC, EMV, PERT, ROI, and CBR to assess the financial impact and provide strategic recommendations for future project investments.

2. Baseline vs. Revised Overview

Original Baselines

- **Scope Baseline:**
 - Fully functional mobile banking app with core banking features
 - Integration with existing banking systems
- **Schedule Baseline:**
 - Planning Phase: 1 month
 - Development Phase: 6 months
 - Testing Phase: 3 months
 - Deployment Phase: 2 months
 - **Total Duration:** 12 months

- **Cost Baseline:**
 - Planning: \$50,000
 - Development: \$300,000
 - Testing: \$100,000
 - Deployment: \$50,000
 - **Total Cost:** \$500,000

Updated Baselines

Due to regulatory changes identified at the 6-month mark, the project scope was expanded to include additional security features.

- **Updated Scope:**
 - Includes enhanced security features to meet compliance requirements
- **Updated Schedule:**
 - Development extended by 2 months
 - **Total Duration:** 14 months
- **Updated Cost:**
 - Development cost increased by \$100,000
 - **Total Cost:** \$600,000

3. ETC & EAC Calculations

Estimate to Complete (ETC)

To calculate ETC, we consider the remaining costs after the first 6 months:

- Remaining Development (including security feature): \$100,000
- Testing Phase: \$100,000
- Deployment Phase: \$50,000

$$\text{Total ETC} = \$100,000 + \$100,000 + \$50,000 = \$250,000$$

Actual Cost (AC)

After 6 months, the following costs have already been incurred:

- Planning Phase: \$50,000
- Initial Development Phase: \$300,000

$$\text{Total AC} = \$50,000 + \$300,000 = \$350,000$$

Estimate at Completion (EAC)

Using the formula:

$$EAC = AC + ETC$$

$$EAC = \$350,000 + \$250,000 = \$600,000$$

This revised EAC reflects the impact of the additional scope and extended development timeline.

4. EMV Analysis

Scenario 1: Without Security Feature

- Probability of threat occurrence: 90%
- Impact if threat occurs: \$80,000
- $EMV = 0.90 \times \$80,000 = -\$72,000$

Scenario 2: With Security Feature

- Probability of threat occurrence: 5%
- Impact if threat occurs: \$6,000
- $EMV = 0.05 \times \$6,000 = -\300

The drastic reduction in Expected Monetary Value (EMV) from -\$72,000 to -\$300 clearly supports the decision to include the security feature. This change significantly reduces the financial and compliance risk exposure, making it a strategically sound investment despite the increase in project cost.

5. PERT Estimation

To validate the revised development schedule, the PERT (Program Evaluation and Review Technique) method was applied using three-point estimation:

- Optimistic (O): 7 months
- Most Likely (M): 8 months

- Pessimistic (P): 9 months

PERT Formula:

$$\text{Expected Duration} = \frac{O + 4M + P}{6}$$

Calculation:

$$\text{Expected Duration} = \frac{7 + 4(8) + 9}{6} = \frac{7 + 32 + 9}{6} = \frac{48}{6} = 8 \text{ months}$$

The expected development duration is **8 months**, which aligns with the revised schedule. This confirms that the updated timeline is realistic and achievable, supporting accurate planning and resource allocation.

6. ROI & Cost-Benefit Analysis

Assumptions

- Revenue and savings are expected to remain stable over the 2-year period following deployment.
- All financial figures are based on estimates provided in the case study.

Total Benefits (Over 2 Years)

- Revenue: \$200,000/year × 2 = \$400,000
- Operational Savings: \$50,000/year × 2 = \$100,000
- Total Benefits = \$400,000 + \$100,000 = \$500,000

Net Profit (Over 2 Years)

- Net Profit = Total Benefits – Total Cost
- Net Profit = \$500,000 – \$600,000 = -\$100,000

Cost-Benefit Ratio (CBR)

- Net Profit = Total Benefits – Total Cost
- Net Profit = \$500,000 – \$600,000 = -\$100,000

Return on Investment (ROI)

Using the formula:

$$ROI = \left(\frac{Net\ Profit}{Investment}\right) \times 100$$

$$ROI = \left(\frac{-100,000}{500,000}\right) \times 100 = -20\%$$

Cost-Benefit Ratio (CBR)

Using the formula:

$$CBR = \frac{Total\ Benefits}{Total\ Costs} = \frac{500,000}{600,000} = 0.83$$

The project yields a negative ROI of -20% and a CBR of 0.83, indicating that the financial return does not justify the investment in purely monetary terms. However, these figures must be weighed against the strategic value of regulatory compliance, risk mitigation, and long-term customer trust.

7. Variance Analysis

The table below summarizes the variances in key metrics:

Metric	Baseline	Revised	Change (Abs)	Change (%)	Notes
Total Project Cost	\$500,000	\$600,000	\$100,000	20%	Increased due to added security scope
Total Duration	12	14	2	16.67%	Extended development
ROI (2-year)	25%	-20%	-45 pp	-	Based on initial investment
Cost-Benefit Ratio (CBR)	1	0.83	-0.17	-17%	Ratio (Benefits/Costs)

The variance analysis highlights a significant shift in both cost and schedule due to the scope change. The ROI dropped by **45 percentage points**, and the CBR fell below 1, indicating reduced financial efficiency. These changes may affect resource allocation and prioritization of future projects. However, the strategic necessity of compliance and risk mitigation justifies the deviation from original baselines.

8. Cash Flow & Break-even Analysis

Cumulative Cash Flow Overview

- Annual Benefit (Revenue + Savings):
 $\$200,000$ (revenue) + $\$50,000$ (savings) = $\$250,000/\text{year}$
- Total Project Cost:
 $\$600,000$

Break-even Calculation

- Payback Period from Go-Live:
 $\$600,000 \div \$250,000/\text{year} = 2.4$ years
- Total Time from Project Start to Break-even:
 14 months (project duration) + 28.8 months (2.4 years) ≈ 3.6 years

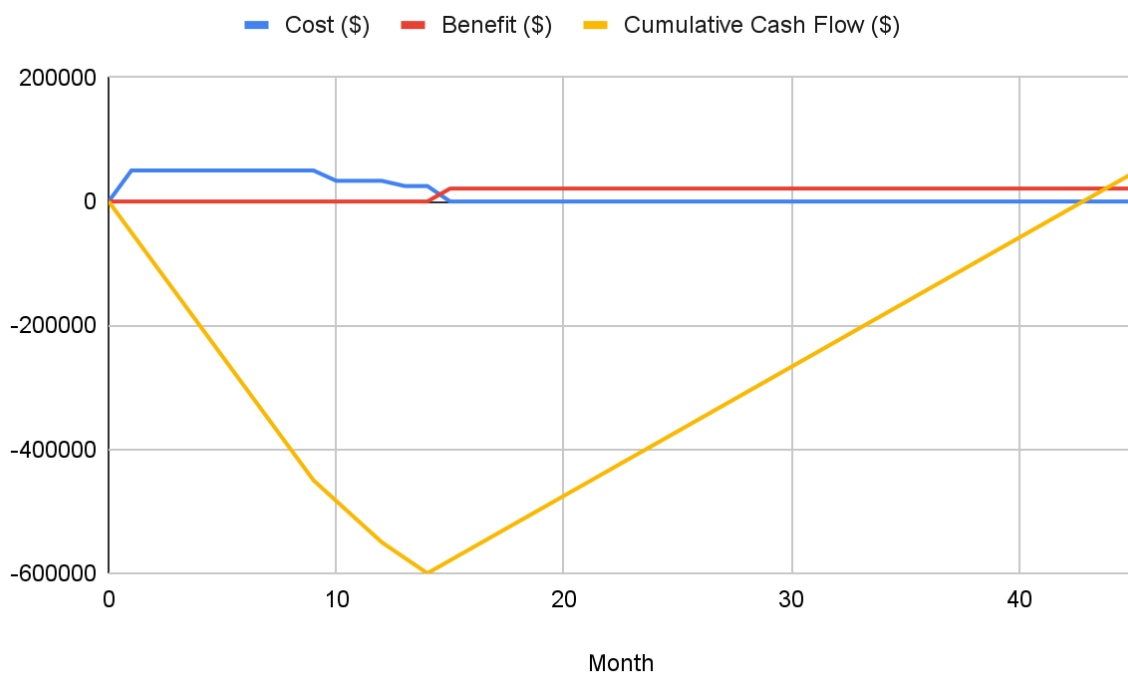


Figure 3: Cumulative Cash Flow & Break-even

The project is expected to reach break-even approximately **3.6 years** after initiation. This exceeds the original 2-year ROI target, indicating a longer recovery period. However, the

extended timeline is justified by the strategic value of compliance, reduced risk exposure, and potential for future scalability and monetization.

9. Recommendations

Short-Term Strategies

- Introduce **premium features** or **tiered service** plans to create new revenue streams.
- Explore strategic partnerships with fintech providers or service aggregators to expand market reach and functionality.

Long-Term Strategies

- Accelerate digital adoption across banking operations to reduce overhead and improve efficiency.
- Streamline internal workflows and automate routine tasks to lower operational costs.
- Apply strict change control procedures to prevent scope creep and maintain budget discipline.
- Strengthen contingency planning to proactively address future regulatory changes and compliance risks.

Final Recommendation

Despite the negative short-term ROI and extended break-even period, the project remains a strategically sound investment. The inclusion of security features ensures regulatory compliance, reduces risk exposure, and builds long-term customer trust. Future projects of similar nature should be evaluated not only on financial metrics but also on their strategic alignment with organizational goals and regulatory landscapes.